



FINAL EXAMINATION SEMESTER III, SESSION 2020/2021

COURSE CODE : FSPP 0034
COURSE : PHYSICS 1I
PROGRAMME : FOUNDATION UTM
DURATION : 3 HOURS
DATE : 11 APRIL 2021

INSTRUCTION TO CANDIDATES:

1. ANSWER ALL QUESTIONS.

WARNING!

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This examination paper consists of (12) printed pages only including this page

QUESTION 1 (20 MARKS)

- a) Determine the temperature of tungsten if it has the same resistivity as iron at 40 °C. Suppose the reference temperature is 20 °C. $T = 233\text{ }^{\circ}\text{C}$

(3 marks)

- b) Figure 1 shows the power rating label on a hair dryer. Based on the information, calculate:

- The peak current it uses. 17.7 A
- The resistance it produces. 19.2 ohm
- The power if the voltage was halved. Assume resistance is constant. 750 W

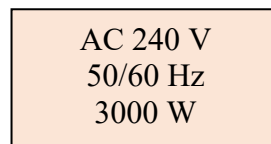


Figure 1

(4 marks)

- c) A copper wire with a diameter of 15 mm has a current density of $9.05 \times 10^4\text{ Am}^{-2}$.

Find:

- The current carried by the copper wire. 16 A
- The number of electrons per hour at any point inside the wire. $3.6 \times 10^{23}\text{ electron/hour}$

(4 marks)

- d) Five identical resistors are connected in a circuit as shown in Figure 2. Deduce the equivalent resistance. $\frac{8}{3} R$

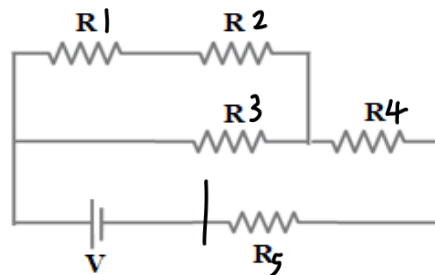


Figure 2

(3 marks)

$$\begin{aligned}
 1) \quad (a) \quad S_T &= S_0 (1 + \alpha (40 - 20)) \\
 &= 9.71 \times 10^{-8} (1 + 0.0645 (20)) \\
 &= 2.22 \times 10^{-7} \Omega \text{m}
 \end{aligned}$$

$$\begin{aligned}
 2.22 \times 10^{-7} &= 5.6 \times 10^{-8} (1 + 0.045 (T - 20)) \\
 3.964 &= 1 + 0.045T - 0.9 \\
 0.045T &= 3.864 \\
 T &= 85.87
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad (i) \quad P &= V_{rms} I_{rms} \\
 3000 &= 240 (I_{rms}) \\
 I_{rms} &= 12.5 \\
 I_0 &= 12.5 \times \sqrt{2} \\
 &= 17.68 \text{ A}
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad V_{rms} &= I_{rms} R \\
 240 &= 12.5 R \\
 R &= 19.2 \Omega
 \end{aligned}$$

$$\begin{aligned}
 (iii) \quad V &= \frac{240}{2} = 120 \text{ V} \\
 I &= \frac{120}{19.2} \\
 &= 6.25 \text{ A} \\
 P &= V_{rms} \times I_{rms} \\
 &= 750 \text{ W}
 \end{aligned}$$

$$\begin{aligned}
 (c) \quad (i) \quad d &= 15 \times 10^{-3} \text{ m}, \quad j = 9.05 \times 10^4 \\
 j &= \frac{I}{A}, \quad A = \pi (7.5 \times 10^{-3})^2 \\
 9.05 \times 10^4 &= \frac{I}{1.767 \times 10^{-4}} \\
 I &= 15.99 \text{ A}
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad Q &= It = 15.99 \times (60 \times 60) = 57600 \text{ Ah}^{-1} \\
 Q &= ne, \quad 57600 = n (1.6 \times 10^{-19}) \\
 n &= 3.6 \times 10^{23} \text{ electrons}
 \end{aligned}$$

$$\begin{aligned}
 (d) \quad R_{12} &= R_1 + R_2 = 2R \\
 \frac{1}{R_{123}} &= \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \\
 \frac{1}{R_{123}} &= \frac{1}{2R} + \frac{1}{R} \\
 \frac{1}{R_{123}} &= \frac{3}{2R} \\
 R_{123} &= \frac{2}{3} R \\
 R_{1234} &= R_{123} + R_4 = \frac{2}{3} R + R = \frac{5}{3} R \\
 R_{12345} &= R_{1234} + R_5 = \frac{5}{3} R + R = \frac{8}{3} R
 \end{aligned}$$

$$\begin{aligned}
 (e) \quad \text{Loop 1} \quad & 12 = I_1(1) + I(6) \\
 & I_1 + 6I = 12 \\
 & \frac{6.4}{1.4} I + 6I = 12 \\
 & 10.57 I = 12 \\
 & I = 1.135 \text{ A} \\
 \text{Loop 2} \quad & 6 = I_2(0.2) + I(6) \\
 & 0.2 I_2 + 6I = 6 \\
 & 0.4 I_2 + 12I = 12 \\
 & I = I_1 + I_2 \\
 & 0.4(I - I_1) + 12I = I_1 + 6I \\
 & 0.4I - 0.4I_1 + 12I - I_1 - 6I = 0 \\
 & 6.4I - 1.4I_1 = 0 \\
 & 6.4I = 1.4I_1 \\
 & I_1 = \frac{6.4}{1.4} I
 \end{aligned}$$

(f) $V = 12V$, $C = 4.4 \times 10^{-6}F$, $R = 2300\Omega$

$$Q = CV = 4.4 \times 10^{-6}(12) \quad \frac{Q}{2} = 2.64 \times 10^{-5}$$

$$= 5.28 \times 10^{-5}C$$

$$Q = Q_0 e^{-\frac{t}{RC}}$$

$$2.64 \times 10^{-5} = 5.28 \times 10^{-5} e^{-\frac{t}{2300(4.4 \times 10^{-6})}}$$

$$\ln(0.5) = \ln\left(e^{-\frac{t}{0.01012}}\right)$$

$$\frac{-25000}{253} = -0.693$$

$$t = 0.0075s$$

2) (a) (i) $N = 200$, $L = 1.84H$, $\Phi = 0.01Wb$, $I = ?$

$$L = \frac{N\Phi}{I}$$

$$1.84 = \frac{200(0.01)}{I}$$

$$I = 1.087A$$

(ii) $l = 0.65m$, $r = 0.06m$, $N = 5000$, $A = \pi(0.06^2) = 0.0113m^2$, $\frac{dI}{dt} = 2.6$

$$M = \frac{\mu_0 N_1 N_2 A}{l} = 7.65 \times 10^{-3}$$

$$\mathcal{E} = -M \frac{dI}{dt}$$

$$= -7.65 \times 10^{-3}(2.6)$$

$$= 0.0199V$$

(iii) $\tau = \frac{L}{R}$

$$0.5 - 2 \times 10^{-3} = 0.498A, \quad t = 3, \quad R = 100\Omega$$

$$0.543 = \frac{L}{100\Omega}$$

$$L = 54.3\Omega$$

$$0.002 = 0.5 e^{-\frac{3}{\tau}}$$

$$4 \times 10^{-3} = e^{-\frac{3}{\tau}}$$

$$\frac{3}{\tau} = 5.52$$

$$\tau = 0.543s$$

(b) $L = 315 \times 10^{-6}H$, $I = 1.2A$

(i) $U = \frac{1}{2}LI^2$

$$= \frac{1}{2}(315 \times 10^{-6})(1.2)^2$$

$$= 2.268 \times 10^{-4}J$$

(ii) $t = 5 \times 10^{-3}$

$$P = \frac{W}{t} = \frac{2.268 \times 10^{-4}}{5 \times 10^{-3}}$$

$$= 0.04536W$$

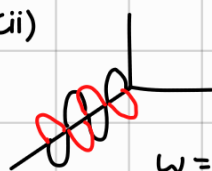
(iii) $\mathcal{E} = -L \frac{dI}{dt}$

$$= -315 \times 10^{-6} \left(\frac{1.2}{5 \times 10^{-3}} \right)$$

$$= 0.0756V$$

(c) (i) transverse wave (ii) $v = f\lambda$, $\frac{3 \times 10^8}{4.5 \times 10^{14}} = 6.67 \times 10^{-9}m$

(ii)



$$E_z = E_0 \sin(kx - \omega t)$$

$$= 4 \sin(1.25 \times 10^6 x - 40\pi t)$$

$$\omega = f\lambda = 40\pi$$

$$\frac{E}{\epsilon_0} = v$$

$$\frac{4}{3 \times 10^8} = B_0$$

$$B_0 = 1.33 \times 10^{-8}$$

$$B_z = 1.33 \times 10^{-8} \sin(1.25 \times 10^6 z - 40\pi t)$$

- e) Determine the magnitude of current in the $6\ \Omega$ resistor shown in Figure 3 if emf ε_1 has an internal resistance of $1\ \Omega$ and emf ε_2 has an internal resistance of $0.2\ \Omega$. **1.13 A**

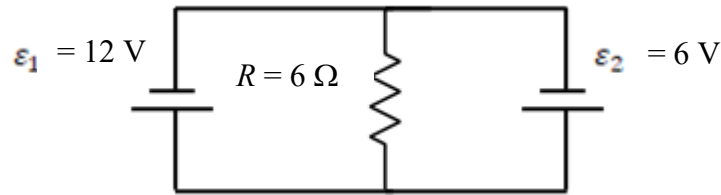


Figure 3

(4 marks)

- f) A battery of 12 V, a $4.4\ \mu\text{F}$ capacitor and $2.3\ \text{k}\Omega$ resistor are connected in series. The battery fully charged the capacitor before it is removed from the circuit. Find the time taken when the charge is half from its maximum value. **7.01 ms**

(2 marks)

QUESTION 2 (20 MARKS)

- a) i. A coil has 200 turns. The self-inductance of the coil is $1.84\ \text{H}$ and the magnetic flux linkage with the coil is $0.01\ \text{Wb}$ when a current flow through the coil. Find the current in the coil. **1.09 A**
- ii. A primary coil 65 cm in length has a radius of 6 cm and 5000 turns. Current is flowing through at a rate of $2.6\ \text{A s}^{-1}$. Calculate the induce e.m.f. in the secondary coil with 70 turns. **19.9 mV**
- ~~iii.~~ The current in an RL circuit drops from $0.5\ \text{A}$ to $2\ \text{mA}$ after 3 s the battery is removed from the circuit. If R is $100\ \Omega$, find the inductance L in the circuit. **54.4 H**

(8 marks)

- b) A $315\ \mu\text{H}$ inductor can carry a maximum current of $1.2\ \text{A}$. Determine:
- i. The amount of energy stored in the inductor when the maximum steady current is flowing. **2.27×10^{-4}**
- ii. The average power of the inductor if the energy stored in (i) is completely discharged in an interval of 5 ms. **4.54×10^{-2}**
- iii. The back e.m.f. induced in the inductor for the same time interval. **7.56×10^{-2}**

- iv. Explain how energy can be stored in an inductor.

the energy is actually stored in the magnetic field generated by the current flowing through inductor

(6 marks)

- c) i. What type of wave is electromagnetic waves?
 ii. A 40 Hz wave is a sinusoidal wave propagating in the z direction with E pointing in the x-direction, and $E_0 = 4.0$ V/m. Write down an expression for electric field and magnetic field.
 iii. Red light from laser pointer has a frequency near 4.5×10^{14} Hz. What is the wavelength of this red light? 667nm

$$i) E = 4\hat{i} \sin(8.38 \times 10^{-7} \hat{z} - 251.33t)$$

$$B = 1.33 \times 10^{-6} \hat{j} \sin(8.38 \times 10^{-7} \hat{z} - 251.33t)$$

(6 marks)

QUESTION 3 (20 MARKS)

- a) A pencil is placed at a distance 15.0 cm from a spherical mirror. A virtual image with linear magnification of 4 is formed.

- i. Calculate the focal length of the mirror. 20 cm

- ii. State the type of the mirror. concave mirror

- iii. Show the image formation using ray diagram.

$$m = \frac{d_i}{d_o} \quad 4 = \frac{d_i}{15}$$

$$d_i = -60$$

$$\frac{1}{f} = \frac{1}{15} - \frac{1}{60}$$

(4 marks)

- b) A ray of laser light is incident on a prism with a refractive index, $n = 1.31$ at a 45° angle to one face as shown in Figure 4.

- i. Calculate the angle at which the light ray emerges from the opposite face. 36.97

- ii. Based on the Figure 4, explain the propagation of light from the prism to the air medium by using Snell's law.

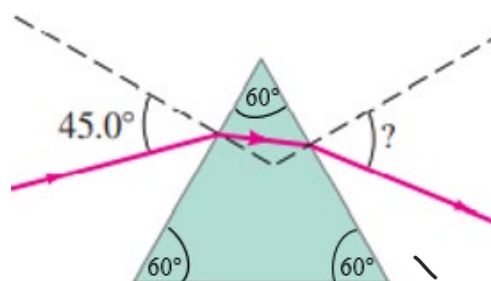
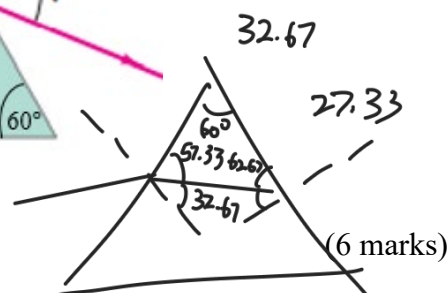


Figure 4



(6 marks)

3) (b) (i)

$$n = \frac{\sin i}{\sin r}$$

$$n = \frac{\sin i}{\sin r}$$

$$1.31 = \frac{\sin 45}{\sin r}$$

$$r = 32.67^\circ$$

$$1.31 = \frac{\sin i}{\sin 27.33}$$

$$i = \sin^{-1}(1.31 \times \sin 27.33)$$

$$= 36.97^\circ$$

$$90^\circ - 32.67^\circ = 57.33^\circ$$

$$180^\circ - 60^\circ - 57.33^\circ = 62.67^\circ$$

$$90^\circ - 62.67^\circ = 27.33^\circ$$

(ii) $n_{\text{air}} = 1$, $n_{\text{glass}} = 1.31$, since $n_{\text{air}} < n_{\text{glass}}$, the light ray bends towards normal, angle $i >$ angle r if light ray enters glass prism from air because $n_1 \sin i = n_2 \sin r$.

$$\begin{aligned} \text{(c) (i)} \quad \frac{1}{p} &= (n-1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \\ &= (1.6 - 1) \left(\frac{1}{-6} + \frac{1}{5} \right) \\ &= 0.02 \text{m} \end{aligned}$$

$$p = 50 \text{cm}$$

$$\text{(ii)} \quad m = -\frac{d_i}{d_o} = \frac{-(-75)}{30} = 2.5$$

$$\frac{1}{p} = \frac{1}{d_i} + \frac{1}{d_o}$$

$$0.02 = \frac{1}{30} + \frac{1}{d_i}$$

$$d_i = -75 \text{cm}$$

$$\begin{aligned} \text{(d) (i)} \quad \frac{1}{p_1} &= \frac{1}{d_o} + \frac{1}{d_i} \\ \frac{1}{6} &= \frac{1}{10} + \frac{1}{d_i} \end{aligned}$$

$$d_i = 15 \text{cm}$$

$$m = \frac{-15}{10}$$

$$= -1.5$$

$$d_{o2} = 40 - 15$$

$$= 25 \text{cm}$$

$$\begin{aligned} \frac{1}{p_2} &= \frac{1}{d_{o2}} + \frac{1}{d_{i2}} \\ \frac{1}{12} &= \frac{1}{25} + \frac{1}{d_{i2}} \end{aligned}$$

$$d_{i2} = 23.08 \text{cm}$$

$$m = \frac{-23.08}{25}$$

$$= -0.92$$

$$\begin{aligned} \text{(ii)} \quad m &= -1.5(-0.92) \\ &= 1.38 \end{aligned}$$

lens 1 : magnified, inverted, real

lens 2 : diminished, inverted, real

Image : magnified, upright, real

- c) A pencil is placed 30.0 cm from a plastic lens which has a concave surface P of radius 6.0 cm and convex surface Q of radius 5.0 cm. Assume that the value of refractive index of plastic lens is 1.60. Determine:
- The focal length of the lens. 50 cm
 - The magnification of the image formed if the pencil is facing the surface Q of the lens. 2.5

(4 marks)

- d) An experimental setup consists of two coaxially convex lenses, A and B with the focal lengths of 6.0 cm and 12.0 cm, respectively. Sample A is placed 10.0 cm in front of lens A. If the separation between two lenses is 40.0 cm, determine:
- The position of the final image. 23.08 cm
 - The magnification of the final image formed by the combination of the two lenses.
(For each lens, state the characteristic of the image formed). 1.38

(6 marks)

APPENDIX 1

List of constant

$$p = +1.60 \times 10^{-19} \text{ C}$$

$$e = -1.60 \times 10^{-19} \text{ C}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$$

$$\varepsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{N}^{-1} \text{m}^{-2}$$

$$g = 9.80 \text{ m s}^{-2}$$

$$c = 3 \times 10^8 \text{ m s}^{-1}$$

$$m_e = 9.11 \times 10^{-31} \text{ kg}$$

$$k = 9.0 \times 10^9 \text{ N m}^2 \text{C}^{-2}$$

$$h = 6.63 \times 10^{-34} \text{ J s}$$

$$N_A = 6.022 \times 10^{23}$$

$$1 \text{ u} = 1.6605 \times 10^{-27} \text{ kg} = 935 \text{ MeV}/c^2$$

$$1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$$

$$\text{Resistivity of iron at } 20^\circ \text{C: } 9.71 \times 10^{-8} \Omega \text{m}$$

$$\text{Resistivity of tungsten at } 20^\circ \text{C: } 5.6 \times 10^{-8} \Omega \text{m}$$

$$\text{Resistivity of platinum at } 20^\circ \text{C: } 10.6 \times 10^{-8} \Omega \text{m}$$

$$\text{Temperature coefficient of iron, } \alpha: 0.0651^\circ \text{C}^{-1}$$

$$\text{Temperature coefficient of tungsten, } \alpha: 0.045^\circ \text{C}^{-1}$$

$$\text{Temperature coefficient of platinum, } \alpha: 0.00397^\circ \text{C}^{-1}$$

APPENDIX 2

List of Formula

$$E = k \frac{Q}{r^2}$$

$$F = k \frac{Q_1 Q_2}{r^2}$$

$$U = k \frac{qQ}{r}$$

$$V = k \frac{Q}{r}$$

$$W = q V$$

$$\Delta U = q \Delta V$$

$$\Delta V = -Ed$$

$$F = q v B \sin \theta$$

$$B = \frac{\mu_0 I}{2\pi d}$$

$$F = ILB \sin \theta$$

$$U = \frac{1}{2} LI^2$$

$$P = IV$$

$$\rho_T = \rho_o [1 + \alpha(T - T_o)]$$

$$j = \frac{I}{A}$$

$$I = \frac{Q}{t}$$

$$M = \frac{\mu_o N_1 N_2 A}{l}$$

$$L = \frac{\mu_o N^2 A}{l}$$

$$I = I_{max} (1 - e^{-t/\tau}) = \frac{V}{R} (1 - e^{-t/\tau})$$

$$Q = Q_o e^{-t/\tau}$$

$$Q = Q_o (1 - e^{-t/\tau})$$

$$I = I_o e^{-t/\tau}$$

$$\varepsilon = -L \frac{dI}{dt}$$

$$\tau = \frac{L}{R}$$

$$\varepsilon = -\frac{d\phi_B}{dt}$$

$$\frac{V_S}{V_P} = \frac{N_S}{N_P}$$

$$v = f\lambda$$

$$E = Bv$$

$$k = \frac{2\pi}{\lambda}$$

$$P = \frac{1}{f}$$

$$m = -\frac{d_i}{d_o} = \frac{h_i}{h_o}$$

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

$$\frac{1}{f} = (n-1) \left[\frac{1}{r_1} + \frac{1}{r_2} \right]$$

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$d \sin \theta = m\lambda, m = 0, 1, 2, 3 \dots$$

$$d \sin \theta = \left(m + \frac{1}{2} \right) \lambda, m = 0, 1, 2, 3 \dots$$

$$Q = \Delta m c^2$$

$$KE = \frac{hc}{\lambda} - W_o$$

$$\left| \frac{dN}{dt} \right|_o = \lambda N_o$$

$$\left| \frac{dN}{dt} \right| = \left| \frac{dN}{dt} \right|_o e^{-\lambda t}$$

$$\lambda = \frac{0.693}{T_{1/2}}$$

$$\varepsilon = \left| -M \frac{dI}{dt} \right|$$

$$I = \frac{\varepsilon}{R}$$

$$\varepsilon = NB\omega A$$

$$\varepsilon = Blv$$

$$B_o = \frac{E_o}{c}$$

APPENDIX 3

Nuclide	Binding energy per nucleon, E (MeV)
$^{95}_{38}\text{Sr}$	8.74
$^{139}_{54}\text{Xe}$	8.39
$^{235}_{92}\text{U}$	7.60

Nuclide	Binding energy per nucleon, E (J)
$^{144}_{56}\text{Ba}$	1.3341×10^{-12}
$^{90}_{36}\text{Kr}$	1.3864×10^{-12}
$^{235}_{92}\text{U}$	1.2191×10^{-12}

Nuclide	Mass, m (u)
$^{235}_{92}\text{U}$	235.04392
$^{141}_{55}\text{Cs}$	140.91963
$^{93}_{37}\text{Rb}$	92.92157
^1_0n	1.00867
^1_1H	1.007825
^2_1H	2.014102
^3_1H	3.016049
^3_2He	3.01605